

SUKHOTHAI, Thailand

WMO: 483720

Lat: 17.1069N

Lon: 99.8003E

Elev: 164

StdP: 14.61

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	62.2	65.4	54.9	64.5	69.0	58.8	74.5	72.9	7.8	75.6	7.4	77.1	1.0	N/A	0.318

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	18.4	102.1	80.6	99.6	80.6	97.4	80.3	83.9	95.5	83.1	93.5	82.5	92.0	3.1	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.4	164.2	88.4	80.7	160.7	87.5	80.1	157.2	86.8	48.3	95.9	47.3	93.6	46.5	92.3	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.9	9.4	7.5	58.2	103.3	3.7	3.0	55.5	105.5	53.3	107.2	51.2	108.9	48.5	111.0		
			56.9	85.3	4.4	1.1	53.7	86.1	51.1	86.8	48.6	87.4	45.4	88.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.2	77.4	80.6	84.1	88.2	87.4	85.3	84.3	83.8	84.2	83.8	81.4	77.7
	DBStd	4.50	3.62	3.02	4.03	3.76	4.11	2.43	2.47	1.85	2.05	2.23	2.95	3.91
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	1	1	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12116	850	856	1058	1145	1161	1058	1063	1049	1025	1049	943	859
	CDD65	6641	385	436	593	695	696	608	598	584	575	584	493	394
	CDH74	76951	3234	4458	7323	9664	9289	7396	7022	6637	6654	6794	5091	3389
Wind	WSAvg	35026	1128	1983	3839	5633	5077	3295	2879	2521	2686	2788	1990	1207
	WSAvg	1.6	0.7	1.0	1.7	2.4	2.1	2.1	2.4	1.9	1.3	1.0	1.1	1.0
	PrecAvg	47.6	0.4	0.5	1.2	2.5	7.4	6.0	5.6	7.3	9.6	5.9	1.1	0.4
	PrecMax	69.1	2.1	2.8	6.3	6.3	15.0	10.4	10.2	10.4	14.4	10.8	4.4	2.8
	PrecMin	33.9	0.0	0.0	0.0	0.3	1.3	1.9	3.2	4.0	5.2	1.3	0.0	0.0
Precipitation	PrecStd	8.3	0.6	0.7	1.5	1.6	4.3	2.2	1.8	2.1	2.7	2.6	1.2	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.6	95.8	101.5	104.6	106.5	98.7	96.9	93.8	95.3	95.1	93.8	93.2
		MCWB	76.1	76.7	75.7	81.6	83.2	80.8	80.9	81.7	82.3	81.5	80.2	77.8
	2%	DB	89.6	93.5	98.6	102.0	102.9	95.4	93.8	91.8	93.2	93.1	92.0	90.7
		MCWB	75.2	77.0	77.4	81.1	82.0	81.0	80.4	80.8	81.4	80.9	79.3	76.1
	5%	DB	88.0	91.8	96.5	99.9	99.6	93.2	91.8	90.5	91.6	91.6	90.5	88.8
		MCWB	74.8	77.0	77.6	81.2	81.5	81.0	80.0	80.5	81.2	80.6	78.6	75.9
	10%	DB	86.2	89.8	94.4	97.8	96.5	91.2	90.0	89.1	89.9	90.1	89.0	86.6
		MCWB	74.3	76.4	77.7	81.0	81.3	80.6	79.6	80.2	80.8	80.3	78.1	75.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.5	81.7	82.3	85.0	85.4	83.4	82.7	83.5	83.6	83.5	82.5	80.2
		MCDB	87.4	90.2	93.6	98.9	98.7	91.7	91.4	91.0	92.2	91.4	90.3	87.9
	2%	WB	78.3	80.2	81.0	83.6	84.2	82.5	81.7	82.3	82.7	82.4	81.0	79.0
		MCDB	86.0	88.8	92.6	97.3	97.4	90.9	90.1	89.3	90.3	89.7	88.6	86.7
	5%	WB	77.4	79.2	80.1	82.6	83.1	81.8	81.0	81.6	81.9	81.7	80.3	77.7
		MCDB	84.4	88.0	91.6	96.0	94.9	89.9	88.8	88.2	89.0	88.7	87.6	85.0
	10%	WB	76.2	78.2	79.1	81.6	82.2	81.2	80.4	80.8	81.4	81.0	79.5	76.5
		MCDB	82.7	86.7	90.1	94.3	92.8	89.0	87.6	87.2	88.1	87.7	86.1	83.6
Mean Daily Temperature Range	5% DB	MDBR	18.0	20.0	19.9	18.4	15.6	12.4	11.3	11.1	11.9	12.4	14.9	17.5
		MCDBR	19.8	21.9	22.9	20.4	20.3	15.0	13.9	13.0	14.2	14.7	16.4	19.2
		MCWBR	7.7	8.4	7.5	6.3	6.0	5.1	5.2	5.2	5.2	4.6	5.4	6.9
	5% WB	MCDBR	16.5	19.3	19.9	19.1	17.2	13.3	12.4	12.2	13.3	13.5	15.0	16.9
		MCWBR	7.4	8.2	7.4	6.9	6.4	5.3	5.1	5.4	5.3	5.0	5.4	6.3
Clear-Sky Solar Irradiance	taub		0.476	0.573	0.678	0.628	0.500	0.443	0.432	0.440	0.479	0.548	0.469	0.470
	taud		2.095	1.863	1.652	1.808	2.196	2.380	2.410	2.373	2.251	2.007	2.205	2.135
	Ebn at Noon		255	236	216	229	257	269	272	272	263	239	256	253
	Edh at Noon		49	64	82	70	47	39	38	39	44	55	43	46
All-Sky Solar Radiation	RadAvg		1478	1653	1728	1902	1762	1586	1485	1454	1485	1484	1512	1424
	RadStd		76	70	104	97	139	122	100	82	84	88	82	61

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	+0.25	N/A	N/A	+185

Nomenclature: See separate page